Enterprise Network Documentation

1. Introduction

This network has been designed to represent a comprehensive infrastructure for a large enterprise, taking into account the essential requirements of various departments to ensure efficient and secure operations. This documentation aims to provide a comprehensive overview of the network design, its components, and connection details.

2. General Network Structure

The network consists of five floors, each designed to host specific departments and functions, reflecting the organizational distribution of the enterprise:

Floor	Main Departments / Function
First	Offices and Customer Service, Warehouses, Cashier, Reception, General Archive, Staff Affairs, Healthcare, Eligibility Office
Second	Offices, Employee Lounge, Commercial Department, IT Department
Third	Administrative Offices, Service Manager, Human Resources, Marketing, Archive
Fourth	Administrative Offices, Manager's Office
Fifth	**First Half:** Monitoring Office (Cameras), Cybersecurity Department. **Second Half:** Server Room (Data Center)

3. Devices and Technologies Used

The network primarily relies on Cisco devices and includes a wide range of equipment to ensure connectivity and required functionalities:

3.1. Core Network Devices

Device	Description
Core Switch Layer 3	Core switches on each floor, plus two main (primary and backup) core switches
Switch Layer 2	Branch switches on floors and in the server room
Router	Two main (primary and backup) routers for Internet connectivity
Access Point (AP)	Wireless access points to provide Wi-Fi coverage

3.2. End Devices and Servers

Category	Devices
User Devices	Personal Computers (PCs), Laptops, Tablets, Smartphones, IP Phones
Printers	Wired and wireless printers
Surveillance	Surveillance Cameras, Monitors (TVs)
Servers	Web and Email server (WEB-Email-server), DNS server, DHCP server, unnamed server (future)
Other	Cooling fans in server rooms, Coffee Machines

4. Connection and Interconnectivity Design

4.1. Core Switches

- A Core Switch Layer 3 is present on each floor (Layer1, Layer2, Layer3, Layer4, Security Section(Layer5), CenterRoom(Layer5)).
- All floor core switches are connected to two main Core Switch Layer 3 devices: MainSwitch (primary) and AlternativePath1 (backup) to ensure high availability and fault tolerance.
- EtherChannel technology is used between some core switches to increase bandwidth and flexibility.

4.2. Routers

- There are two main routers: internet1 (primary) and internet2 (backup).
- Both MainSwitch and AlternativePath1 are connected to internet1 and internet2.
- EtherChannel technology is used between the main core switches and routers to increase bandwidth and flexibility.

4.3. Data Center

- Contains two Layer 2 switches:
  - DMZ(CenterRoom(Layer5)): Dedicated to the DMZ zone, directly connected to internet1 and internet2.
  - CenterRoom(Layer5): Dedicated to the remaining servers (DNS, DHCP, Server 3), connected to the main core switches MainSwitch and AlternativePath1.
- A Web and Email server (WEB-Email-server) is located in the DMZ zone.

4.4. Wireless Network

- Wireless connectivity is provided via Access Points (APs) distributed across different floors.
- APs are connected directly to Layer 2 switches or Core Switches.

4.5. Cameras

- All cameras in the network (wired and wireless) are connected to a dedicated VLAN for surveillance.
- Cameras are connected to specific ports on Layer 2 switches or Core Switches.

5. Floor and Device Details

5.1. First Floor (Layer1)

- **Core Switch:** Core Switch Layer 3 named Layer1.
- **Branch Switches:** Two Layer 2 Switches named S1.1 and S1.2.
- **Access Points:** AP S1.1-1.1 (connected to S1.1) and AP S1.2-2.1 (connected to S1.2).
- **Departments and Devices:**
  - **Warehouse Office:** PC, IP Phone, Printer (connected to S1.1).
  - **Cashier Office (Fund):** PC, IP Phone, Printer (connected to S1.2).
  - **Reception and Customer Service:** 5 PCs, 5 IP Phones, Printer, Tablet (connected to S1.2).
  - **Shared Offices (one hall):** General Archive, The Staff Affairs, Healthcare, Eligibility Office. PCs, IP Phones, two shared printers (connected to S1.1).

- **Cameras:** A surveillance camera connected to S1.2 on port Fa0/10, named C Layer1(S1.2-2.1).
- **Branch Switches to Core Switch Connection:** S1.1 and S1.2 connect to Layer1 via ports Fa0/1 and Fa0/2 respectively, and port Fa0/24 on the branch switches.
- **APs to Switches Connection:** APs connect to branch switches on port Fa0/16.

5.2. Second Floor (Layer2)

- **Core Switch:** Core Switch Layer 3 named Layer2.
- **Branch Switches:** None.
- **Access Points:** AP Layer2-2.1 (connected to Layer2 on port Fa0/16) and AP-Lounge(Layer2-2.2) (connected to Layer2 on port Fa0/17).
- **Departments and Devices:**
    - **Commercial Department Office:** 3 PCs, 3 IP Phones, Printer (wireless).
    - **IT Office:** 2 PCs (one with IP Phone), Printer, Laptop, Tablet.
    - **Employee Lounge:** Coffee machine (wireless), laptops, smartphones, and tablets connect wirelessly.
- **Cameras:** A surveillance camera in the lounge named C Layer2(Lounger2.2-2.1), connected wirelessly to AP-Lounge(Layer2-2.2).
- **Devices to Core Switch Connection:** Devices connect directly to Layer2 starting from port Fa0/1.

5.3. Third Floor (Layer3)

- **Core Switch:** Core Switch Layer 3 named Layer3.
- **Branch Switches:** None.
- **Access Points:** AP Layer3-3.1 (connected to Layer3 on port Fa0/16).
- **Departments and Devices:**
    - **Service Manager Office:** PC, IP Phone, Printer (wireless).
    - **Human Resources (HR) Department:** 2 PCs, 2 IP Phones, Printer.
    - **Marketing Department:** PC, IP Phone, 2 Laptops (wireless), Tablet.
    - **Archive Department:** 2 PCs (one with IP Phone), Printer (wireless), Smartphone.
- **Cameras:** A camera named C Layer3(AP3.1-1.1) connected directly to AP Layer3-3.1.
- **Devices to Core Switch Connection:** Devices connect directly to Layer3 starting from port Fa0/1.

5.4. Fourth Floor (Layer4)

- **Core Switch:** Core Switch Layer 3 named Layer4.
- **Branch Switches:** Three Layer 2 Switches named S4.1, S4.2, and S4.3.
    - Branch switches connect to Layer4 via port Fa0/1 on Layer4 to port Fa0/24 on the branch switches.

- **Access Points:**
  - AP S4.1-1.1 (connected to S4.1 on port Fa0/16).
  - AP S4.2-2.1 (connected to S4.2 on port Fa0/16).
  - AP S4.3-3.1 (connected to S4.3 on port Fa0/16).
- **Departments and Devices:**
  - **Branch Switch S4.1:**
    - **Manager's Office Manager:** PC, IP Phone, Printer (wireless).
    - **Meeting Room:** TV, Tablet (connect wirelessly to AP S4.1-1.1).
    - **Manager Office:** PC, IP Phone, Laptop (wireless), Smartphone.
    - **Manager's Secretary:** PC, IP Phone, Printer (wireless) (on the same VLAN as Manager Office).
    - **Public Relations Office:** 3 PCs, 3 IP Phones, shared Printer.
  - **Branch Switch S4.2:**
    - **Deputy Office:** PC, IP Phone, Smartphone.
    - **Deputy's Secretary:** PC, IP Phone, Printer (wireless) (on the same VLAN as Deputy Office).
    - **Director General of Finance and Accounts:**
      - Director: PC, IP Phone.
      - Accounts Officer: PC, IP Phone.
      - Checks and Bank Officer: PC, IP Phone.
      - Auditor: PC, IP Phone.
      - Two wireless printers.
  - **Branch Switch S4.3:**
    - **Oversight Office:** 2 PCs, 2 IP Phones, Printer (wireless).
    - **Technical Department Manager Office:** PC, IP Phone, Printer (wireless), Tablet.
    - **Statistics Office:** PC, IP Phone, Printer (wireless).
- **Cameras:** A basic camera connected to S4.3 via cable Fa0/10, named C Layer4(S4.3-3.1).
- **End Device Connection:** All end devices connect to branch switches starting from port Fa0/1.

5.5. Fifth Floor (First Half - Security and Monitoring Department)

- **Core Switch:** Core Switch Layer 3 named Security Section(Layer5).
- **Branch Switches:** Two Layer 2 Switches named S5.1 (for monitoring) and S5.2 (for cybersecurity).
- **Access Points:** AP S5.2-2.1w (connected to S5.2 on port Fa0/16).
- **Departments and Devices:**
  - **Monitoring Department:** PC, IP Phone, Printer (wired). Connected to S5.1.
  - **Cybersecurity Department:** PC, IP Phone, PC (without IP Phone), 2 Laptops (wired), Printer (wireless), 2 TVs (wireless), Tablet. Connected to S5.2.

- **Cameras:** A camera named C CY connected to S5.2 on port Fa0/10. In addition to 15 cameras distributed throughout the company (one camera per floor, two in the server room, one in the cybersecurity department, one external camera, one at the main entrance, one at the back entrance, two in the parking lot, one in the warehouse, one on the fifth floor).
- **Branch Switches to Core Switch Connection:** S5.1 and S5.2 connect to Security Section(Layer5) via ports Fa0/1 and Fa0/2 respectively, and port Fa0/24 on the branch switches.

5.6. Fifth Floor (Second Half - Data Center)

- **Core Switch:** Core Switch Layer 3 named CenterRoom(Layer5).
- **Branch Switches:** Two Layer 2 Switches named DMZ(CenterRoom(Layer5)) and CenterRoom(Layer5).
- **Routers:** internet1 and internet2.
- **Devices and Servers:**
  - **DMZ(CenterRoom(Layer5)):** Connects to internet1 (port Gig0/1 to Gig0/1/3) and internet2 (port Gig0/2 to Gig0/1/3). Connected to WEB-Email-server (port Fa0/1) and DMZ FAN (port Fa0/15).
  - **CenterRoom(Layer5):** Connects to MainSwitch (port Gig0/1 to Gig1/0/15) and AlternativePath1 (port Gig0/2 to Gig1/0/15). Connected to DNS (port Fa0/1), DHCP (port Fa0/2), Server 3 (port Fa0/3), and CenterRoom(Layer5) FAN (port Fa0/15).
- **Cameras:** Two wired cameras named C CenterRoom1 (port Fa0/10) and C CenterRoom2 (port Fa0/11).

6. Main Core Switches and Routers Interconnectivity

6.1. Floor Core Switches to MainSwitch and AlternativePath1 Connection

Floor Core Switch	Connection to MainSwitch (Floor Core Switch Port -> MainSwitch Port)	Connection to AlternativePath1 (Floor Core Switch Port -> AlternativePath1 Port)
Layer1	Gig0/1 -> Gig1/0/11	Gig0/2 -> Gig1/0/11
Layer2	Gig0/1 -> Gig1/0/12	Gig0/2 -> Gig1/0/12
Layer3	Gig0/1 -> Gig1/0/13	Gig0/2 -> Gig1/0/15
Layer4	Gig0/1 -> Gig1/0/14	Gig0/2 -> Gig1/0/14

Floor Core Switch	Connection to MainSwitch (Floor Core Switch Port -> MainSwitch Port)	Connection to AlternativePath1 (Floor Core Switch Port -> AlternativePath1 Port)
Security Section(Layer5)	Gig0/1 -> Gig1/0/16	Gig0/2 -> Gig1/0/16
CenterRoom(Layer5)	Gig0/1 -> Gig1/0/15	Gig0/2 -> Gig1/0/15

6.2. MainSwitch and AlternativePath1 to Routers Connection

- **MainSwitch:**
  - Gig1/0/1 -> internet1 (Gig0/0/0)
  - Gig1/0/2 -> internet1 (Gig0/0/1)
  - Gig1/0/3 -> internet2 (Gig0/0/0)
  - Gig1/0/4 -> internet2 (Gig0/0/1)
  - (EtherChannel used between MainSwitch and routers)
- **AlternativePath1:**
  - Gig1/0/1 -> internet1 (Gig0/1/0)
  - Gig1/0/2 -> internet1 (Gig0/1/1)
  - Gig1/0/3 -> internet2 (Gig0/1/0)
  - Gig1/0/4 -> internet2 (Gig0/1/1)

7. Proposed VLAN and IP Address Segmentation

To ensure security, efficiency, and ease of management, the network will be segmented into logical VLANs, with appropriate IP ranges allocated to each. The 192.168.0.0/16 network will be used as an example for the overall enterprise IP range.

VLAN ID	VLAN Name	Description	Proposed IP Range (Example)
10	Administration	Administrative offices (Third and Fourth Floors, Manager's Office, Public Relations, Deputy Office, Director General of Finance and Accounts, Oversight, Statistics)	192.168.10.0/24
20	Customer Service	Reception and Customer Service (First Floor)	192.168.20.0/24
30	Warehouse & Cashier	Warehouse office, Cashier office (First Floor)	192.168.30.0/24
40	Shared Offices	General Archive, Staff Affairs, Healthcare, Eligibility Office (First Floor)	192.168.40.0/24

VLAN ID	VLAN Name	Description	Proposed IP Range (Example)
50	Commercial Department	Commercial Department (Second Floor)	192.168.50.0/24
60	IT Department	Information Technology Department (Second Floor)	192.168.60.0/24
70	Human Resources	Service Manager, Human Resources (Third Floor)	192.168.70.0/24
80	Marketing	Marketing Department (Third Floor)	192.168.80.0/24
90	Surveillance	All cameras in the network and monitoring office (Fifth Floor)	192.168.90.0/24
100	Cybersecurity	Cybersecurity Department (Fifth Floor)	192.168.100.0/24
110	Internal Servers	DNS, DHCP, Server 3 (Server Room)	192.168.110.0/24
120	DMZ	Web and Email server (Server Room)	192.168.120.0/24
130	Wireless Network	All wireless devices connected to APs (can be further segmented later)	192.168.130.0/24
140	IP Phones	All IP Phones in the network	192.168.140.0/24

Notes on VLAN Segmentation:

- **Cameras:** All cameras will be on a single VLAN (VLAN 90) for easy management and centralized monitoring.
- **Servers:** DMZ servers will be separated from internal servers in different VLANs for increased security.
- **Wireless Network:** Initially, all wireless devices can be placed on a single VLAN (VLAN 130). It can later be segmented into sub-VLANs based on departments or user types if necessary.
- **IP Phones:** A separate VLAN (VLAN 140) can be dedicated to IP Phones to ensure Quality of Service (QoS).
- **Cybersecurity:** The Cybersecurity Department (VLAN 100) will have access and monitoring privileges over all other VLANs to enable them to perform their security tasks effectively.

8. Honeypot Server Addition Plan

To enhance network security and detect intrusion attempts, a Honeypot Server will be added.

- **Proposed Location:** The Honeypot server can be placed in VLAN 100 (Cybersecurity), VLAN 110 (Internal Servers), or even VLAN 120 (DMZ) depending on the objective of the Honeypot.
    - **If the goal is to detect internal attacks:** It is preferable to place it in VLAN 110 or VLAN 100.
    - **If the goal is to detect external attacks:** It is preferable to place it in VLAN 120 (DMZ).
- **Initial Recommendation:** It is recommended to place it in VLAN 100 (Cybersecurity) to be under the direct supervision of the Cybersecurity Department, with access only by them, allowing them to effectively analyze threats without exposing real systems to risk.

9. Standards

To ensure network quality and stability, the following standards must be adhered to:

- **Cisco Standards:** Utilize Cisco best practices and recommendations for device configuration.
- **IEEE Standards:** Adhere to IEEE standards for Ethernet and wireless networks (e.g., 802.3 and 802.11).
- **Security Standards:** Implement strict security policies, including access control, encryption, and intrusion detection/prevention systems.
- **Naming Conventions:** Adhere to naming conventions for devices, ports, and VLANs for ease of management and maintenance.
- **Documentation:** Regularly update this documentation to reflect any changes to the network.